Observation of thermionic emission of electrons, space-charge-limited-current in a vacuum diode and the Franck-Hertz experiments on Hg and Ne

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A vacuum diode with varying heating and acceleration voltages was investigated. The Langmuir-Schottky law, describing the space-charge-limited-current (SCLC) in a vacuum, was checked using heating voltages of 6.5 V and 7.5 V and acceleration voltages between −500 V and 500 V. In the SCLC region the data showed a power law behaviour with an exponent of $(1,3500 \pm 0,0097)$ for 6.5 V heating voltage and $(1,3709 \pm 0,0083)$ for 7.5 V. The deviation from the expected $\frac{3}{2}$ exponent were discussed. The Richardson equation was checked using heating voltages between 5 V and 7 V at an acceleration voltage of 500 V and the metal work function of the tungsten cathode was determined to (6.266 ± 0.069)eV. The deviation from the value found in literature was discussed. It was observed that the diode works like a rectifier. Furthermore, the Franck-Hertz experiment was reconstructed with neon and mercury filled tubes. The excitation energies were determined to (19.9 ± 1.4)eV and (2.08 ± 0.14)eV for the neon filled tube and (4.91 ± 0.16)eV for the mercury filled tube. These energies were compared to the emitted light observed and with transitions in the Grotrian diagram.

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1 Introduction

Before semiconductors were pure enough to be used in computational electronics vacuum diodes and triodes were used. Later they became obsolete because semiconductor devices evolved and provided smaller size and better performance. Nevertheless, the importance of vacuum diodes for the early days of electronic computing and communication was huge. Therefore it is important to understand how the vacuum diode works and to check formulas describing its behaviour, which is done in the following experiments. The Franck Hertz experiment was first introduced by James Franck and Gustav Hertz in 1914. It showed that atoms can only be excited with discrete energies. In the case of the Franck Hertz experiment these atomic excitations were observed using the kinetic energy of accelerated electrons. The atoms electrons get excited into a higher energy state, where they later spontaneously emit light when recombining to the previous energy level. The experiment shows energy levels are discrete -meaning that only electrons with a specific kinetic energy can cause this effect. This leads to the characteristic lighting phenomena for certain atoms, e.g. neon.
This knowledge is the cornerstone for the understanding of modern physics with its quantised energy levels. Trickle down effects of this understanding include applications in the lighting industry as well as in laser related technology. For example, it can be used for target marking or plays an essential role for under water communicating submarines. In the study at hand the foundations for the understanding of this are

laid by reconstructing the experiment that was conducted by James Franck and Gustaf Hertz in 1914.

2 Experiment

Free electrons can be generated via thermionic emission. If a current flows through a wire the wire heats up due its resistance. With increasing temperature, more and more electrons have enough energy to overcome the work function and escape the wire into a vacuum. The emission current density of those free electrons is described by the Richardson equation

$$j = CT^2 \mathrm{e}^{-\frac{W}{k_B T}} \tag{1}$$

where W is the work function of the cathode, C the Richardson constant and $k_B = 1.38 \times 10^{-23}\,\mathrm{JK}^{-1}$ [2] the Boltzmann constant.

2.1 Characteristics of a vacuum diode

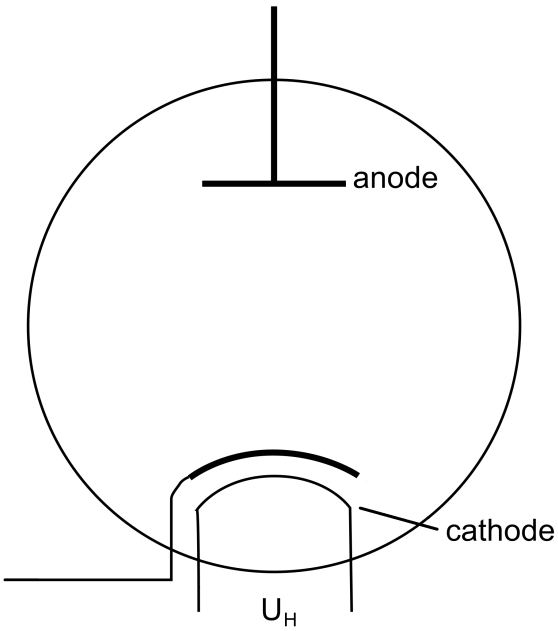

Fig. 1: Schematic setup of a vacuum tube. The electrons generated at the cathode through thermionic emission caused by U_H are accelerated towards the anode by a voltage U_a. [3]

Using a vacuum diode like Figure 1 depicts the anode current was recorded for different heating voltages U_H and acceleration voltages (between anode and cathode). The results are shown in Figure 2. For the heating voltages 6.5 V and 7.5 V

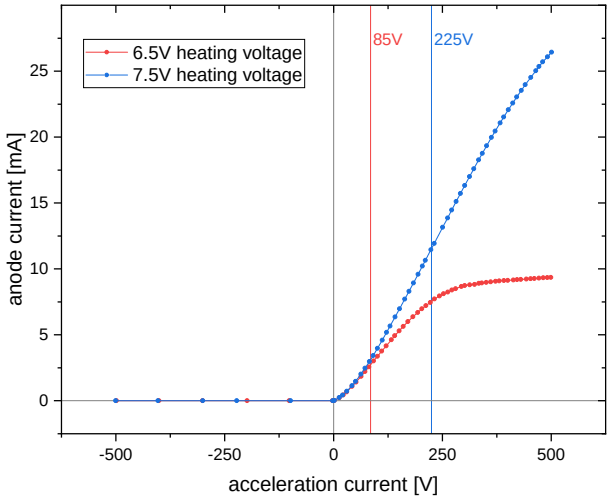

Fig. 2: Characteristic curve of a vacuum diode for 6.5 V and 7.5 V as heating voltages of the hot cathode and for acceleration voltages between −500 V and 500 V. One can see that the diode works like a rectifier and the current reaches saturation depending on the heating voltage.

the current at the anode was measured for acceleration voltages between −500 V and 0 V in steps of 100 V, meaning that the hot cathode in Figure 1 was at a higher potential than the anode. Since the charge carriers are electrons the diode works as a rectifier in this voltage range. If a voltage in the opposite direction is applied a measurable current emerges. To correctly describe the current flow in a vacuum diode one has to consider the space charge that surrounds the cathode. The free electrons generated by thermionic emission partially counter the potential difference of the cathode and the anode. The space-charge-limited-current (SCLC) in a vacuum diode can be described by the Langmuir-Schottky law

$$j = \frac{4}{9}\epsilon_0 \sqrt{\frac{2e}{m}} \frac{U^{\frac{3}{2}}}{d^2} \tag{2}$$

where U is the acceleration voltage between cathode and anode, e the elementary charge, m the electron mass, ϵ the vacuum permittivity and d the distance between the two electrodes.

2.1.1 Langmuir-Schottky region

Figure 2 shows that Langmuir-Schottky law (2) only holds in a delimited voltage range. Since Equation 2 is a strictly convex function only the part of the data that shows similar behaviour is estimated to obey this law. The regions start at 0 V and the estimated ends are respectively depicted by the vertical lines in Figure 2.

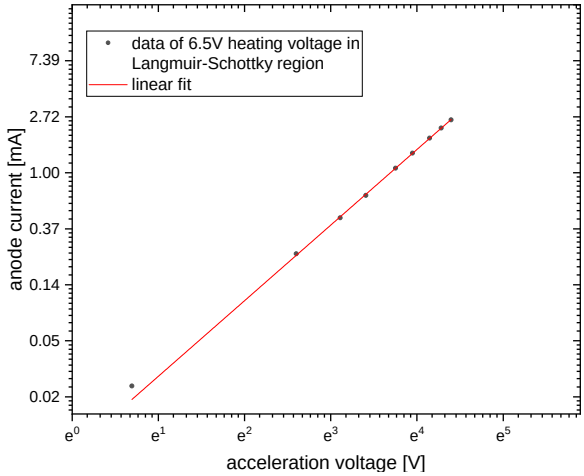

Fig. 3: The data in the estimated Langmuir-Schottky region for 6.5 V heating voltage. One can recognize a linear behaviour.

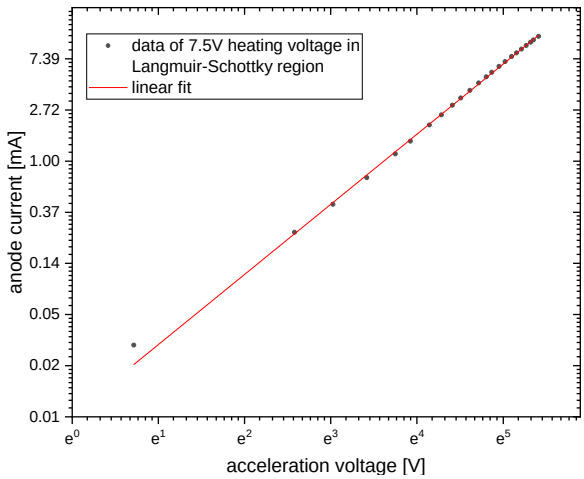

Fig. 4: The data in the estimated Langmuir-Schottky region for 7.5 V heating voltage. One can recognize a linear behaviour.

The data points belonging to the estimated SCLC region are plotted double logarithmically in Figure 3 and Figure 4. A linear fit should show the power of the power law (2) as its slope. The obtained slopes are $(1,3500 \pm 0,0097)$ for 6.5 V and $(1,3709 \pm 0,0083)$ for 7.5 V. The results do not coincide with the ideal value of $\frac{3}{2}$, even within range of their error. One can see that Figure 2 shows that the current is only exponentially increasing at low voltages and then shows linear behaviour in U like a normal resistor would. This suggests that the assumption of a vacuum is not valid and there still occur some scattering events with remaining gas molecules/atoms, influencing the current and making the use of the Langmuir-Schottky equation not entirely correct. This is also supported by the obtained slope values. Due to the scattering events one would expect a behaviour between ohm's law (linear behaviour) and the $U^{\frac{3}{2}}$ power law of (2).

Furthermore, at a higher heating voltage more electrons can escape the cathode. The fraction of electrons interacting with remaining gas atoms decreases relative to the number of electrons being undisturbed resulting in a higher slope found for the 7.5 V heating voltage.

2.1.2 Saturation region

If the acceleration voltage is too high, all generated electrons are immediately drawn off towards the anode. Even with increasing voltage the current doesn't increase much since there are not more charge carriers available. Due to more electrons capable of escaping the cathode the saturation current for the 7.5 V heating voltage is higher and occurs at higher acceleration voltages. Therefore, saturation was not reached by the measurement but the beginning of the saturation region can be seen. The saturation current is supposedly described by the Richardson equation (1). The currents at an acceleration voltage of $\approx$ 500 V were recorded for varying heating voltages between 5 V and 7 V, assuming that saturation was reached. The results are shown in Figure 5.

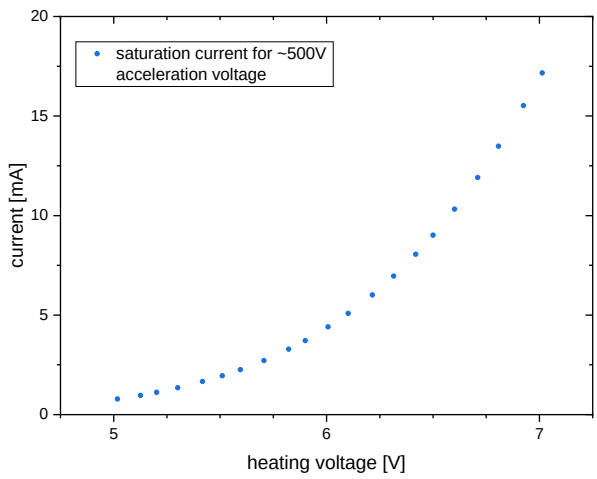

Fig. 5: The currents of the saturation region at an acceleration voltage of $\approx$ 500 V for heating voltages between 5 V and 7 V. One can see an exponential behaviour in accordance with the Richardson equation.

To check if the Richardson equation holds for the present data one needs the temperature of the cathode. The used cathode was made out of

tungsten, for which the approximation

$$T = 1616\,\mathrm{K} + 97\,\frac{\mathrm{K}}{\mathrm{V}} U_H \tag{3}$$

is used to determine its temperature depending on the heating voltage U_H. The resulting equation is

$$\ln \frac{j}{T^2} = \ln C - \frac{W}{k_\mathrm{B}} \frac{1}{T} \tag{4}$$

Since $j \propto I$ the value I/T^2 is plotted logarithmically against $\frac{1}{T}$ in Figure 6. From Equation 4

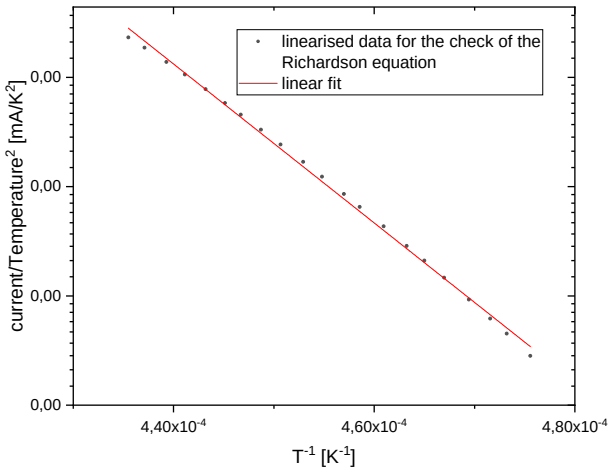

Fig. 6: Linearised plot according to the Richardson equation to determine the work function of the tungsten cathode. The data can be described by a linear fit.

one can obtain the metal work function of wolfram. For the present data it is calculated to be (6.266 ± 0.069)eV. This value does not coincide with the value of 4.60eV [4] found in literature. This may be caused by the acceleration voltage which was too low for the higher heating voltages to reach saturation. In Figure 2 one can already see, that saturation for 7.5 V heating voltage was not reached. Therefore, at higher heating voltage one underestimates the current and therefore the exponent was overestimated, resulting in a higher metal work function. Also the approximation of Equation 3 could be an error sources. Despite that one can at least see that the values are of the same magnitude. This shows that the experiment at hand is not a realistic example for electron emission in vacuum, implying that the Richardson equation might be a bit over idealized in comparison to the possible experimental realization.

2.1.3 Contact voltage

The contact voltage U_c is the voltage that needs to be applied to the diode to set the current to 0. This is reflected in a shift of the voltage in (2) according to $U \to U - U_c$. This shift can be found by plotting $I^{\frac{2}{3}}$ against U and determining the intersection of a linear fit with the x-axis. The plots are shown in Figure 7 For 6.5 V one finds $U_c = -(2.82 \pm 0.53)$V and for 7.5 V $U_c = -(5.64 \pm 0.94)$V. The values don't coincide within range of their error. The linear fit parameters depend on the range of the data used (the data estimated to obey the Langmuir-Schottky law). For the different heating voltages the errors discussed in subsubsection 2.1.1 have different impact. For 7.5 V a lot more data points were estimated to be in the SCLC region. The data at higher voltages reduce the slope and move the intersection with the x-axis to the left. Therefore, one can assume that the difference in the values is mainly caused by the different data ranges used and the different influence of the errors discussed before.

2.2 Franck-Hertz experiment

In a Franck-Hertz tube like Figure 8 depicts, electrons are emitted at a cathode via thermionic emission, drawn away by a voltage U_1 and then accelerated through a gas by a potential difference U_2 towards a grid. They are then slowed down by a counter voltage so only electrons with sufficient energy reach the anode. From the previous findings the detected current is expected to increase with increasing acceleration voltage. However, in the experiment the detected current shows recesses at certain voltages (Figure 9,Figure 10). One can also see that they are characteristic for the gas used in the diode. This phenomenon suggests that the accelerated electrons regularly interact with the atoms of the gas filling. Furthermore, the recessions indicate that the gas atoms only accept energies in a certain amount, the quantized levels. In the following, the energies of the excited atoms were determined for the the diode filled with neon and with mercury.
Configuration of voltages:
The higher the voltage U_1 the higher the overall emission current. It has to be set high enough

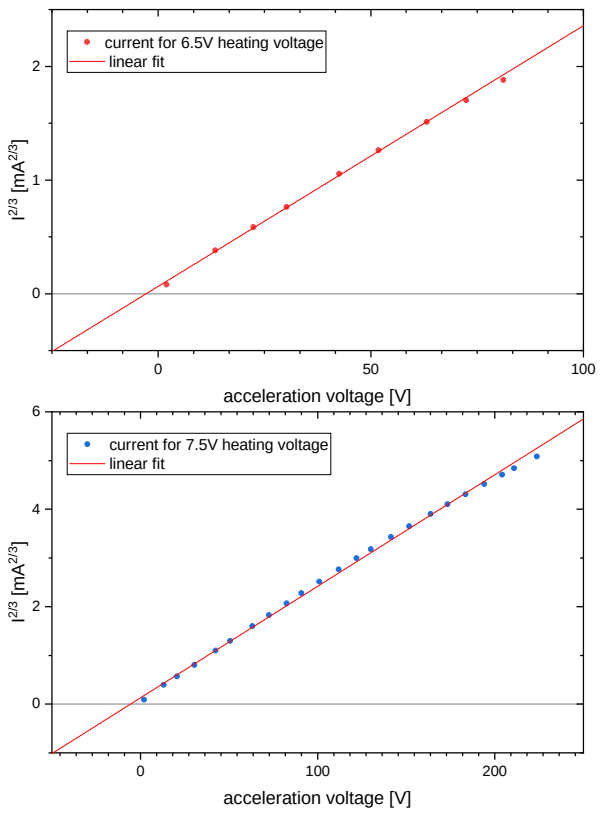

Fig. 7: The recorded current I plotted as $I^{\frac{2}{3}}$ against the acceleration voltage for the heating voltages 6.5 V and 7.5 V. A linear behaviour can be observed. The intersection of the linear fit with the x-axis gives the contact voltage.

to cause a detectable current but must not cause clipping. The breaking voltage U_3 is used to emphasise maxima and minima.

2.2.1 Neon filling

For neon one used an acceleration voltage that was increased from $U_3 = 0\,\mathrm{V} - 80\,\mathrm{V}$. The documentation of the measurement can be seen in Figure 9. In Table 1 the readoff points of the maxima are listed. The differences between latter ones are the same as between the minima. Accordingly, the voltage recesses occur at the points at which most of the electrons have a kinetic energy of $\Delta E = (19.9 \pm 1.4)\mathrm{eV}$. A view on the Grotrian diagram Figure 11 suggests this corresponds to the transition of the ground level to the $2p^5 3p^3 P_0$ level. In addition, smaller maxima can be seen. Especially, after the third maximum two of them are easily visible Figure 9. Again, assuming these to be proportional to the energy difference in the excited atoms, the energy gap

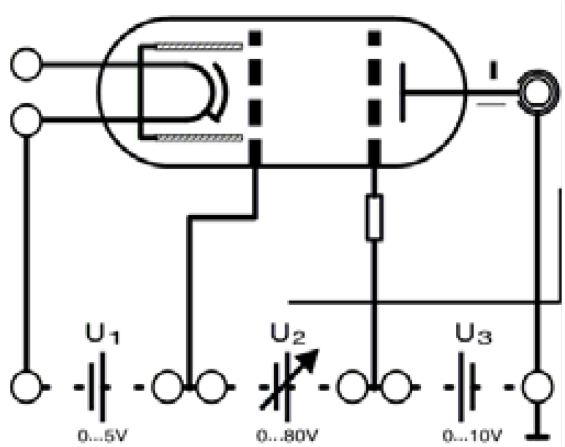

Fig. 8: The basic setup for the Franck-Hertz experiment. The tube consists of 4 electrodes. Free Electrons are generated via thermionic emission at the cathode. They are then drawn away by the voltage U_1. They are then accelerated by the voltage U_2 and transported through the Hg/Ne gas towards the grid G. Passing electrons are then slowed down by the breaking voltage U_3. Electrons with high enough energy reach the anode A and cause a current I_A. [1]

was calculate to $\Delta E = (2.08 \pm 0.14)\mathrm{eV}$. Their smaller size indicates that the transition is less likely than the excitation from the ground level to the $2p^5 3p^3 P_0$ level. The light that one was able to see during the experiment was of orange colour. It can be conjectured that this was due to the emission of photons resulting from the atom falling back to the level of lower energy. This observation matches the theory, that the atoms are excited from the ground level with an energy of 18.9 eV which corresponds to the big maxima seen in Figure 9. From this energy level the electrons fall down, emitting a photon which can be seen as the orange colour. From the lower level they can be excited again with an energy of 2.11 eV depicted by the small maxima.

Table 1: evaluation of the energy levels of neon. The right column contains the energy difference to the next maximum and the average value.

maximum	acceleration voltage [V]	energy difference [eV]
1	18.92	18.3
2	37.22	20.59
3	57.81	20.83
4	78.64	avg.: 19.9 ± 1.4

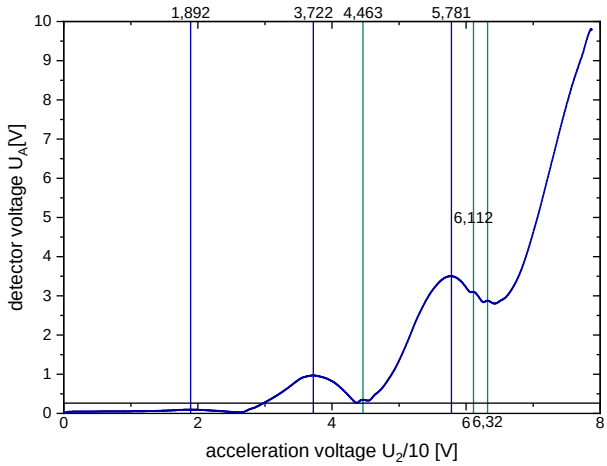

Fig. 9: The characteristic curve for neon obtained by the Franck-Hertz experiment. One can see recesses in the voltage which indicate that the electrons interact with the neon atoms and excite them. Besides the big maxima there occur also small maxima.

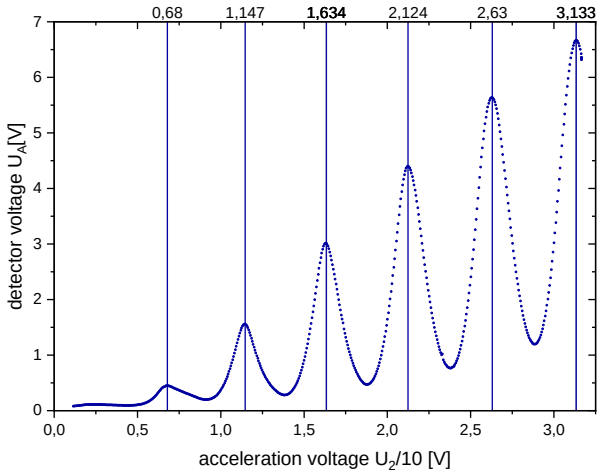

Fig. 10: The characteristic curve for neon obtained by the Franck-Hertz experiment. One can see recesses in the voltage which indicate that the electrons interact with the mercury atoms and excite them.

2.2.2 Mercury filling

At first, part of the mercury in the tube was vaporised via heating. Then, the experiment carried out for neon is repeated for mercury. This time the voltage was increased from $U_3 = 0\,\mathrm{V} - 30\,\mathrm{V}$. The same pattern evolves. It can be seen that the energy gaps are smaller for mercury Figure 10. It was found that the energy gaps were of the size $\Delta E = (4.91 \pm 0.16)$eV Table 2. This can be seen in Figure 10. However, this time no smaller maxima can be seen. The Grotrian diagram of mercury reflects this; from the $E = 4.89\,\mathrm{eV}$ state the atom can only fall down into the ground state. Also, no light in the visible spectrum can be seen as the drop to the ground state emits light in the ultra-violet spectrum.

Table 2: evaluation of the energy levels of mercury. The right column contains the energy difference to the next maximum and the average value.

maximum	acceleration voltage [V]	energy difference [eV]
1	6.8	4.67
2	11.47	4.87
3	16.34	4.9
4	21.24	5.06
5	26.3	5.03
6	31.33	avg.: 4.91 ± 0.16

3 Summary

The experiment fell into two parts. In the first part the characteristics of a vacuum diode are determined. For the electron current in vacuum the Langmuir-Schottky formalism was tested. It was found that the residual pressure in the diode was the main cause of systematic deviation from the Langmuir-Schottky law. Furthermore, the Richardson equation was found to be insufficient for the present with respect to error sources. The work function of the tungsten cathode was overestimated, mainly because saturation was not reached and the saturation current was underestimated. The second part of the experiment comprised the implementation of latter knowledge about the vacuum diode in the Franck-Hertz experiment. It was shown that the diode filled with gas differs fundamentally from the vacuum diode. The existence of regular recesses in the detected electron flow depending on the acceleration voltage suggested that the atomic energy absorption is quantized. For neon, light phenomena were experienced. They were explained by the transition from the excited state to a lower energy state during which a photon is emitted. Also for mercury recesses were detected but the photons emitted by the transition were attributed to the ultra violet spectrum making them not visible for the human eye. This was confirmed by calculation.

4 Appendix

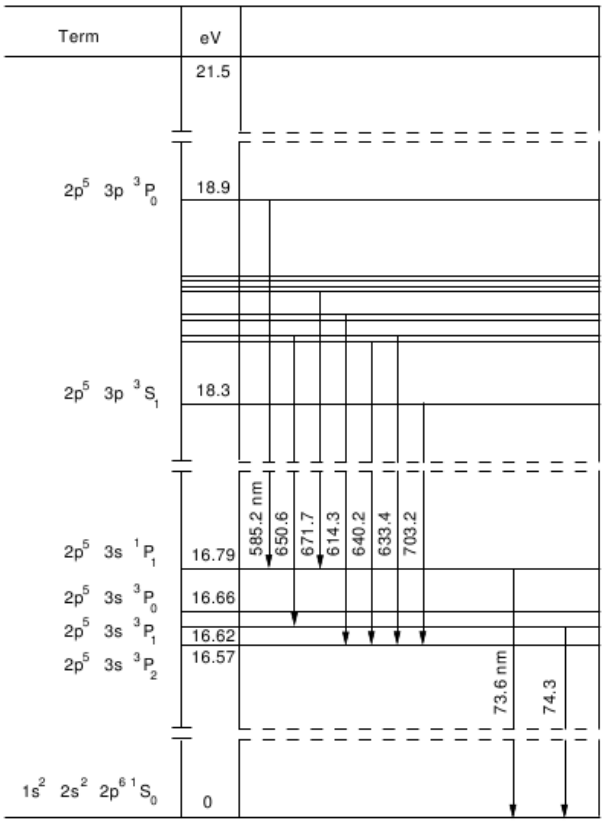

Fig. 11: Energy levels of Neon with allowed transitions and their respective energy.

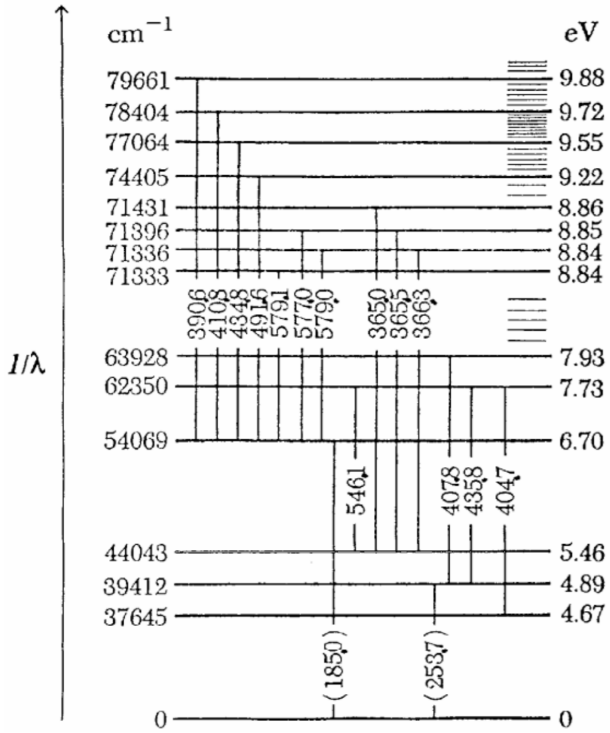

Fig. 12: Energy levels of mercury with allowed transitions and their respective energy.

References

[1] Manual for the C49 experiment of the basic internship, Institute of Physics of the University of Würzburg, `https://www.physik.uni-wuerzburg.de/fileadmin/physik-grundpraktikum/_imported/fileadmin/11016800/Modulbeschreibungen/C-Modul/Versuch_49_ueberarbeitet.pdf` (cropped), last visited September 25, 2020

[2] `https://physics.nist.gov/cgi-bin/cuu/Value?k`, last visited September 25, 2020

[3] `https://commons.wikimedia.org/wiki/File:Vacuum_diode.svg` (edited), last visited September 25, 2020

[4] `https://www.fh-muenster.de/ciw/downloads/personal/juestel/juestel/chemie/Austrittsarbeit.pdf`, last visited September 25, 2020